MAHMOUD YOUNES

Flutter Developer

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PROFESSIONAL SUMMARY

Flutter Developer and Computer Science graduate (Zagazig University, 2025) with hands-on experience building cross-platform mobile applications for Android and iOS using Flutter and Dart. Skilled in Clean Architecture, BLoC state management, REST and GraphQL API integration, and Firebase. Experienced in on-device machine learning integration (TensorFlow Lite), CI/CD automation, and delivering freelance AI solutions for clients. Committed to clean code, SOLID principles, responsive user interfaces, and cross-platform mobile application performance optimization.

TECHNICAL SKILLS

Mobile Development: Flutter, Dart, Cross-Platform Mobile Apps, Responsive User Interfaces, Debugging, Performance Optimization
State Management: BLoC / Cubit, Provider, Riverpod
Backend & APIs: RESTful APIs, GraphQL, Dio, Firebase (Auth, Firestore, Storage), Supabase, DeepSeek API
Databases & Storage: SQLite (sqflite), Hive, Shared Preferences
Architecture: Clean Architecture, MVVM, Repository Pattern, OOP, SOLID Principles
AI / ML Integration: TensorFlow Lite, YOLOv8, Roboflow, On-Device Inference
CI/CD & DevOps: GitHub Actions, Fastlane, Firebase App Distribution, Flutter Flavors, Git & GitHub
Code Quality: Unit Testing, Widget Testing, Freezed, build_runner, json_serializable
Tools: Android Studio, VS Code, Postman, Figma, Adobe XD

PROJECTS

Exam Zone — AI-Powered Educational Platform — Graduation Project

Flutter | Dart | Firebase Auth | Firebase Storage | DeepSeek API — github.com/sayodaa/exam-zone

- Built a cross-platform educational app that reduces the time teachers need to create and correct exams through AI-powered question generation and automated grading via the DeepSeek API
- Implemented separate teacher and student account systems using Firebase Authentication for role-based educational management
- Automated generation of essay and multiple-choice questions with automated grading capabilities

Photo Analysis — On-Device AI Artifact Recognition — Freelance Project

Flutter | Dart | YOLOv8 | TensorFlow Lite — github.com/Mahmoud-Younes/photo_analysis

- Delivered a complete mobile AI application for a client, covering the full cycle from model training to on-device integration
- Trained a custom YOLOv8 detection model to classify 15 distinct artifact types, achieving 96.5% accuracy (mAP@50)
- Optimized and converted the model to TensorFlow Lite for fully offline, on-device inference without internet dependency
- Implemented multi-object recognition to identify several artifacts in a single image with per-artifact confidence scores

Store App — E-Commerce App with Clean Architecture & CI/CD — Personal Project

Flutter | Dart | BLoC | GraphQL | Freezed | GitHub Actions | Fastlane

- Architected the app with feature-first Clean Architecture (presentation / domain / data) using the repository pattern and dependency injection for scalability and testability
- Integrated GraphQL with typed queries and mutations, pagination, and unified error handling
- Managed state with BLoC and built immutable models using Freezed, Equatable, and build_runner code generation
- Debugged and optimized application performance through efficient state management and reduced widget rebuilds
- Wrote unit and widget tests, enforced project-wide lints, and automated builds, release signing, and distribution via GitHub Actions, Fastlane, and Firebase App Distribution with dev/prod flavors

Additional Projects

- **Shop App** — E-commerce app consuming live REST APIs with loading states, pagination, and responsive UI
- **Social App** — Firebase-based social app with authentication and real-time data sync

EDUCATION

Bachelor of Computer Science and Information Systems — Zagazig University, Faculty of Computers and Information

Cairo, Egypt

Oct 2021 – Jul 2025

CERTIFICATIONS

- Flutter Store App: GraphQL, BLoC, Freezed & Clean Architecture
- The Complete Flutter & Dart Development Course
- Flutter Payment Integration: Stripe, PayPal & More

LANGUAGES

Languages: Arabic (Native), English (Conversational)